



Effects of exercise training during hemodialysis on cardiac baroreflex sensitivity

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Key words

hemodialysis patients – cardiorespiratory capacity – baroreflex sensitivity – baroreflex effectiveness index – exercise training

Abstract. Background: Arterial baroreflex sensitivity (BRS) evaluation has been increasingly used as an index of cardiac autonomic control. Cardiac autonomic dysfunction leading to depressed BRS has been associated with an increased risk of ventricular arrhythmias and sudden death in patients with chronic kidney disease (CKD) on hemodialysis (HD). **Aim:** The purpose of this study was to investigate the effects of an exercise training program during hemodialysis on BRS in CKD patients. **Patients and methods:** 43 HD patients participated in the study. They were randomly assigned into either a 7-month exercise training program during HD (Group A: n = 22 patients) or a sedentary control group (Group B: n = 21 patients). Additionally, 20 sex- and age-matched sedentary individuals comprised a healthy control group (Group C). All patients at the beginning and the end of the study underwent a tilt test for evaluation of BRS and an exercise testing with spirometric study for cardiorespiratory capacity estimation. The level of Hb, medications and the HD procedure remained stable during the study. **Results:** At baseline BRS was found to be reduced by 51.5% ($p < 0.05$) and baroreflex effectiveness index (BEI) by 36.4% ($p < 0.05$) in Group A compared with Group C. Initially, all HD patients had also significantly lower exercise time and VO_2 peak than the healthy subjects. After training, Group A showed a significant improvement in BRS by 23.0% ($p < 0.05$), in BEI by 27.0% ($p < 0.05$), in event and ramp count by 35.0% ($p < 0.05$) and 29.0% ($p < 0.05$), respectively as well as in VO_2 peak by 22.4% ($p < 0.05$) and in exercise time by 40.9% ($p < 0.05$). Significant correlations were found between BRS and METs ($r = 0.476$, $p < 0.05$), BRS and VO_2 peak ($r = 0.443$, $p < 0.05$), BEI and METs ($r = 0.492$, $p < 0.05$), BEI and VO_2 peak ($r = 0.467$, $p < 0.05$), event count and VO_2 peak ($r = 0.715$, $p < 0.01$), event count and exercise time ($r = 0.799$, $p < 0.01$), in Group A at the end of the study. **Conclusions:** Our results indicate that HD patients had considerably reduced cardiorespiratory capacity and impaired cardiac BRS compared to healthy se-

dentary individuals. Importantly, exercise training during HD yielded a marked increase of the indices representing baroreflex activity in association to the improvement of their functional capacity.

Introduction

Increased cardiovascular morbidity and mortality are known complications in patients with chronic kidney disease (CKD) for several reasons [Herzog 2003]. Among them, cardiac autonomic dysfunction is common in CKD patients on hemodialysis (HD) leading to arrhythmias and episodes of intradialytic hypotension and also complicating with sudden cardiac death [Deligiannis et al. 1999, Shoji et al. 2004]. Baroreflex sensitivity (BRS) is a widely used marker of cardiac autonomic control evaluating the interaction between cardiac sympathetic and vagal activity [La Rovere 2000]. Specifically, depressed values of BRS indicate an increased cardiac sympathetic activity [La Rovere 2000]. During the last decade measurement of BRS has gained considerable interest, since impairment in the baroreflex control of heart rate has been associated with an increased propensity for cardiac mortality and sudden cardiac death in patients with myocardial infarction, chronic kidney disease and diabetes [Johansson et al. 2007, La Rovere et al. 1998, Pikkujamsa et al. 1998]. Moreover, there have been few studies supporting that BRS was depressed in CKD patients with hypertension as well as on HD [Johansson et al. 2005, Lazarous et al. 1973]. HD patients are continuously under an uremic milieu, resulting in vascular calcification and arterial stiffness which are associated with reduced BRS [Chesterton et al. 2005].

Many studies have demonstrated enhanced BRS and baroreflex effectiveness index (BEI) after an exercise training program in patients

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with COPD, Type 2 diabetes and coronary artery disease leading to a reduction of cardiac risk [Costes et al. 2004, Iellamo et al. 2000, Loimaala et al. 2003]. However, there has been no study assessing the effects of an exercise program on BRS in HD patients. The only favorable effect of exercise training in cardiovascular autonomic modulation in HD patients by increasing the heart rate variability index was described by Deligiannis et al. [1999]. The main question of this study was if an exercise training program during dialysis is feasible to improve the baroreflex activity in HD patients who are under the strenuous effect of uremic toxins. Accordingly, another goal was to determine if there is a relationship between cardiac autonomic and functional capacity benefits of training.

Patients and methods

Patients

From the 60 HD patients who volunteered to participate in the study, 50 patients were judged to be eligible according to established criteria. All of them were on HD for at least 6 months. Patients were excluded from the study if they had any of the following criteria: unstable hypertension, congestive heart failure (Grade > II according to NYHA), cardiac arrhythmias ($N \geq III$ according to Lown), recent myocardial infarction, unstable angina, diabetes mellitus, active liver disease or previous established causes of syncope.

26 of them (Group A) were randomly assigned to participate in a 7-month exercise training program during their HD sessions. The remaining 24 patients (Group B) were asked to continue their usual lifestyle and served as the patients' control group. Moreover, 20 sex- and age-matched healthy sedentary volunteers participated in the study and served as controls (Group C). Informed consent was obtained from all subjects according to the guidelines approved by the Aristotle University Ethical Committee.

Method

All patients were examined at the beginning and after a 7-month period on a non-

dialysis day. The healthy subjects underwent all examinations just once only at the beginning of the study. On the day of the measurements, all subjects attended the laboratory at least 2 hours after a light meal and having abstained from consuming alcohol and caffeine-containing beverages for at least 12 h before the measurements. Then, they underwent complete physical examination and blood tests. Moreover, tilt test and spirometric study during graded treadmill exercise testing were performed. During the 7-month study the level of Hb as well as the medical therapy had remained constant in all patients. Also, the dialysis procedure was kept stable by using the same model of filter and composition of the dialysis solution and keeping dialysis time and frequency constant.

To ensure that the ultimate measurement actually reflects the new autonomic regulation level, patients of Group A repeated the final measurements at least 2 days and no longer than a week after the end of the training program.

Measurement of arterial baroreflex sensitivity

Before each tilt test, subjects rested in a quiet room for 10 min. The measurements were carried out in a quiet, medium-light room with an ambient temperature of 20–24 °C. The subjects were not allowed to talk during the procedures and were instructed not to sleep.

In order to evaluate BRS, the Task Force Monitor (TFM) 3040i device (CNSystem, The brain and heart company, Graz, Austria) was used. The TFM combined oscillometric (OscBP) and continuous plethysmographic blood pressure (cBP) measurement. Noninvasive arterial pressure was also measured on a beat-to-beat basis using the "flying-V" finger cuff of the TFM. A 2-channel ECG was used for monitoring.

The BRS was measured in a 60° head-up position since it was recommended this tilt angle as suitable for activation of baroreflex [Kenny et al. 2000]. Therefore, after a horizontally supine rest period for 5 min, the tilt table was inclined over a period of 30 min on 60° head-up. Afterwards, the subjects were returned to the horizontal supine position and

remained there for other 5 min. During this period, ECG, BP, OscBP and O₂ saturation were continuously recorded.

Data calculations

The BRS value was determined by a spectral analysis of spontaneous fluctuation of systolic blood pressure (SBP) and RR interval (RRI). A linear regression between these two parameters was applied to each individual sequence and the mean slope of the SBP/RRI relationship was calculated and taken as a measure of spontaneous BRS for each period. Only regression lines with a squared correlation coefficient $r^2 > 0.85$ were accepted for analysis. The BRS was the average regression slope for all the linear regressions plotted for accepted baroreflex sequences. Additionally, the following parameters were estimated:

- the baroreflex effectiveness index (BEI), which showed how often concomitant changes (ramps) in SBP and RRI were detected. The BEI was defined as the ratio of the total number of sequences to the total number of identified SBP ramps,
- the ramp count, which consisted of 3 or more consecutive beats in which the SBP values rose or fell together, and
- the event count, which was the number of baroreceptor sequences, characterized by a progressive rise or fall in SBP with a subsequent lengthening or shortening of the RRI for three or more consecutive beats.

From the above parameters the BEI is the most used variable for the estimation of baroreflex function [Di Rienzo et al. 2001, Johansson et al. 2005].

Measurement of cardiorespiratory capacity

After the completion of BRS measurements all subjects rested for 30 min and then underwent a symptom-limited cardiopulmonary test on a treadmill (Trackmaster TM-400, Jas MFG, Carrollton, TX, USA) using the standard Bruce protocol. The patients were monitored continuously using a 12-lead (ECG) throughout the tests (CH-2000,

Cambridge Heart Co, Bedford, MA, USA). Blood pressure (BP) was measured using a sphygmomanometer during the 3rd min of each stage. Tests were terminated when subjects were either exhausted or reached the target heart rate or had symptoms, such as chest pain, dyspnea, or ECG abnormalities appeared. Cardiorespiratory capacity of all subjects was evaluated by a spirometric study. Expired gases were analyzed on a breath-by-breath basis using a spirometer device with fast O₂ and CO₂ analyzers (Quark b², Cosmed, Rome, Italy). Before each test the gas analyzer was calibrated. Validation criteria for a maximal test were lack of increase in VO₂ with an increase in workload, heart rate close to the theoretical age-predicted maximal value and respiratory exchange ratio of more than 1.05.

From the ergospirometric study the following parameters were analyzed: heart rate at peak exercise (HR_{peak}), total exercise time (ET), pulmonary ventilation (VE), metabolic equivalents (METs) and peak oxygen uptake (VO_{2peak}). The VO_{2peak} was considered as the highest VO₂ obtained during a steady state condition with no further increase with increased work rate.

Exercise rehabilitation program

After initial screening, patients of Group A followed a 7-month exercise rehabilitation program during their thrice-weekly dialysis schedule. Many studies reported that hemodialysis patients can be exercised during their hemodialysis sessions which seems to be a more convenient and time-efficient way of training [Kouidi 2002, Parsons et al. 2006]. A major advantage of this method is that it does not create a need for extra time because patients usually dialyze 3 times a week. Therefore, this mode of training is more applicable and preferable from the patient's point of view and the reported compliance with exercise training during hemodialysis is high [Daul et al. 2004]. They were exercised under the supervision of a physician and the responsibility of a physical educator specialized in renal rehabilitation. The prescribed exercise duration was 90 min each time during the first 2 h of the HD session. The duration and intensity of exercise increased progressively dur-

Table 1. Clinical data of subjects (mean $\pm$ SD).

	Group A	Group B	Group C
Subjects (n =)	22	21	20
Male/female	15/7	17/4	15/5
Age (years)	50.05 $\pm$ 13.2	50.52 $\pm$ 14.4	48.10 $\pm$ 9.2
Height (cm)	169.09 $\pm$ 8.8	169.43 $\pm$ 9.2	173.33 $\pm$ 7.4
Weight (kg)	69.98 $\pm$ 11.3	71.33 $\pm$ 8.0	74.95 $\pm$ 9.0
BMI	24.35 $\pm$ 2.7	24.97 $\pm$ 3.2	26.03 $\pm$ 3.5
Months on HD	76.32 $\pm$ 7.0	72.8 $\pm$ 5.3	
Years of kidney disease	10.5 $\pm$ 15.1	8.1 $\pm$ 12.5	
Parameter	N	N	
Causes of chronic kidney disease			
Polycystic kidney disease	6	2	
Tumor	1	0	
Glomerulonephritis	7	10	
Vesica ureteral reflux	1	2	
Urinary tract obstruction	2	1	
Alport disease	0	2	
Nephroscleroses	3	2	
SLE (Systemic lupus erythematosus)	0	1	
Miscellaneous or unknown causes	2	1	
Medications			
Beta receptor blocker	9	7	
Calcium channel antagonist	13	9	
Digitalis	3	2	
Amiodarone	2	1	

ing the study. The exercise routine consisted of a 60 min cycling using specific bedcycles, which were adjusted to each patient's dialysis chair or bed and 30 min strengthening and flexibility exercises. The cycling exercise session started with a 5-min warm-up and terminated with 5 min recovery. During the first week of training patients were cycling passively and then actively. Subjects were exercised at 13 (somewhat hard) of the Borg perceived exertion scale [Borg and Linderholm 1970, Kouidi et al. 2004], in order to be able to complete initially at least 30 min of exercise in a given bout. The duration of the workload was gradually increased over time according to each patient's ability and finally reached 1 h of active cycling. The additional exercise session consisted of sets of repetitions using isotonic and isometric exercises.

The workload was increased progressively by using therabands and applying weights round the limbs. Heart rate and blood pressure were measured every 5 min during the exercise session.

Statistical methods

Data were expressed as mean $\pm$ SD or number of cases. The Kolmogorov-Smirnov test was used to determine the normal distribution of the variables. In order to examine the differences between the independent variables of the research (time and group) a 2-way ANOVA with repeated measures was performed. When there were statistically significant differences, an analysis of simple main effect was administered. Correlation analysis

Table 2. Hematological and biochemical data of patients (mean $\pm$ SD)

	Baseline		After	
	Group A	Group B	Group A	Group B
Hb (gr/dl)	11.7 $\pm$ 1.3	11.9 $\pm$ 1.5	11.8 $\pm$ 1.4	11.7 $\pm$ 1.5
Fe (μ g%)	80.66 $\pm$ 32.9	92.67 $\pm$ 27.5	87.27 $\pm$ 38.3	93.29 $\pm$ 25.4
Glucose (mg%)	96.91 $\pm$ 16.5	93.10 $\pm$ 11.1	97.77 $\pm$ 14.7	94.71 $\pm$ 16.3
K (mEq/l)	5.48 $\pm$ 0.6	5.64 $\pm$ 0.9	5.41 $\pm$ 0.5	5.66 $\pm$ 1.0
Na (mEq/l)	140.45 $\pm$ 4.2	137.71 $\pm$ 3.9	139.27 $\pm$ 2.7	138.38 $\pm$ 5.1
Ca (mg%)	9.08 $\pm$ 0.6	8.72 $\pm$ 0.7	8.84 $\pm$ 0.6	8.58 $\pm$ 0.7
P (μ g%)	6.01 $\pm$ 1.1	5.87 $\pm$ 1.2	6.07 $\pm$ 1.2	5.97 $\pm$ 1.2

Table 3. Baroreflex sensitivity data during the 60° stand up (mean $\pm$ SD)

Gravos	Group A		Group B		Group C
	Baseline	After	Baseline	After	Baseline
BRS (msec/mmHg)	6.1 $\pm$ 1.7	7.6 $\pm$ 1.8**	5.5 $\pm$ 1.8	5.3 $\pm$ 1.6	11.1 $\pm$ 1.5**
BEI (%)	42.9 $\pm$ 8.7	54.6 $\pm$ 9.1*	40.7 $\pm$ 12.0	39.7 $\pm$ 11.0	71.2 $\pm$ 9.2**
Event count	37.1 $\pm$ 12.4	57.7 $\pm$ 24.2**	38.5 $\pm$ 11.9	37.6 $\pm$ 11.0	64.7 $\pm$ 14.4**
Ramp count	127.0 $\pm$ 44.7	180.0 $\pm$ 51.1**	112.9 $\pm$ 40.2	115.5 $\pm$ 39.6	201.4 $\pm$ 49.6**

* $p < 0.05$ between the baseline and the follow-up values, ** $p < 0.05$ between groups A and B, *** $p < 0.05$ between Groups A, B, and C.

between variables was also performed using the Pearson's rank correlation. A correlation obtained was considered as weak ($0.35 < r \leq 0.50$), moderate ($0.50 < r \leq 0.65$), high ($0.65 < r \leq 0.85$) and strong ($0.85 < r \leq 1$). All analyses have been carried out using the Statistical Package for Social Sciences for Windows (version 12.0, SPSS Inc., Chicago, Ill, USA).

Results

Four patients of Group A and 3 of Group B were excluded from the study. A female (Group A) and a male (Group B) refused to complete the final measurements for personal reasons, while another female patient (Group A) was not able to complete the training program because of hypertension crises. Additionally, a female patient (Group B) couldn't repeat the final measurements, due to amputation of her leg. Furthermore, 2 males of Group A and 1 of Group B died of causes unrelated to exercise. There were no musculoskeletal,

cardiovascular or other complications related to the exercise program. Finally, 43 HD patients completed the study: Group A consisted of 22 and Group B of 21 patients. Clinical characteristics of these subjects are presented in Table 1. There were no significant differences between patients and healthy controls regarding gender, age, weight, height and BMI. Instantly, Group C had lower values of resting heart rate by 14.2% ($p < 0.05$), as well as systolic and diastolic blood pressure by 16.1% ($p < 0.05$) and 7.7% ($p < 0.05$), respectively, in comparison with patient groups. No significant differences were observed in the values of Hb, serum urea, creatinine and albumin between the patient groups across the 7-month study (Table 2).

Baseline measurements

Initially, there was no significant difference between the patient groups concerning the BRS values (Table 3). However, baseline

Table 4. Cardiorespiratory efficiency data of subjects (mean $\pm$ SD).

	Group A		Group B		Group C
	Baseline	After	Baseline	After	Baseline
HR rest (bpm)	73.8 $\pm$ 8.4	65.5 $\pm$ 4.1*#	72.6 $\pm$ 7.1	71.5 $\pm$ 7.0	64.6 $\pm$ 5.2**
SBP rest (mmHg)	137.2 $\pm$ 16.7	128.9 $\pm$ 13.2*#	135.7 $\pm$ 16.1	133.7 $\pm$ 14.9	118.00 $\pm$ 12.5**
DBP rest (mmHg)	82.9 $\pm$ 9.3	76.9 $\pm$ 7.9*#	83.57 $\pm$ 7.9	82.4 $\pm$ 7.0	77.00 $\pm$ 9.7**
HR peak (bpm)	133 $\pm$ 21	145 $\pm$ 14*#	134 $\pm$ 10	137 $\pm$ 12	174 $\pm$ 13**
SBP peak (mmHg)	184 $\pm$ 20	171 $\pm$ 18	181 $\pm$ 16	180 $\pm$ 14	166 $\pm$ 14
DBP peak (mmHg)	84 $\pm$ 12	81 $\pm$ 14	83 $\pm$ 11	84 $\pm$ 9	75 $\pm$ 14
ET (min)	6.7 $\pm$ 1.8	8.3 $\pm$ 2.2*#	6.3 $\pm$ 1.6	6.2 $\pm$ 1.7	10.5 $\pm$ 1.6**
VE _{max} (l/min)	44.6 $\pm$ 13.7	59.7 $\pm$ 18.2*#	51.4 $\pm$ 4.9	52.6 $\pm$ 5.7	97.97 $\pm$ 38.0**
VO ₂ peak (ml/Kg/min)	20.5 $\pm$ 5.1	25.1 $\pm$ 6.1*#	21.9 $\pm$ 4.5	20.8 $\pm$ 4.6	34.6 $\pm$ 7.8**
METs	5.8 $\pm$ 1.4	7.1 $\pm$ 1.7*#	6.1 $\pm$ 1.2	5.9 $\pm$ 1.3	9.9 $\pm$ 2.1**

*p < 0.05 between the baseline and the follow-up values, #p < 0.05 between Groups A and B, **p < 0.05 between Groups A, B, and C.

values of BRS and BEI were lower in Group A by 51.5% ($p < 0.05$) and by 36.4% ($p < 0.05$), respectively, compared to Group C. Likewise, event count and ramp count were reduced in Group A by 42.0% ($p < 0.05$) and by 36.0% ($p < 0.05$) respectively, compared to Group C. Similar differences were found at baseline between Groups B and C.

There were no statistically significant differences in baseline measured parameters of the spirergoometric study among patient groups (Table 4). Patients of Group A achieved less exercise time on the treadmill by 40.9% ($p < 0.05$), lower peak heart rate by 21.3% ($p < 0.05$), VE_{max} by 46.2% ($p < 0.05$), VO₂peak by 39.9% ($p < 0.05$) and METs by 40.4% ($p < 0.05$), compared with group C. Similar differences were observed between Groups B and C.

In order to determine the relationship between the level of baroreflex activity and functional capacity at baseline, correlation coefficients were calculated. Highly positive significant correlations were found in all patients between BRS and ET ($r = 0.776$, $p < 0.01$), METs ($r = 0.766$, $p < 0.01$) and VO₂peak ($r = 0.726$, $p < 0.01$), as well as between BEI and ET ($r = 0.710$, $p < 0.01$), METs ($r = 0.738$, $p < 0.01$) and VO₂peak ($r = 0.696$, $p < 0.01$). Further highly positive significant correlations were found between ramp count and VO₂peak ($r = 0.627$, $p < 0.01$), ET ($r = 0.699$, $p < 0.01$), METs ($r = 0.656$, $p < 0.01$)

and VE peak ($r = 0.634$, $p < 0.01$), in all patients. In addition, there were moderate significant correlations between event count and VO₂peak ($r = 0.528$, $p < 0.01$), ET ($r = 0.569$, $p < 0.01$), METs ($r = 0.509$, $p < 0.01$) and VE_{peak} ($r = 0.596$, $p < 0.01$) in all patients. There were inverse significant correlations found in group C between age and BRS and BEI ($r = -0.582$, $p < 0.01$ and $r = -0.847$, $p < 0.01$, respectively). There was no correlation found between BRS and age, duration of CKD or duration of HD, in patient groups.

Follow-up measurements

All indices of cardiac baroreflex activity were found to be improved after exercise training in Group A (Table 3). Specifically, there was a significant increase in BRS by 23.0% ($p < 0.05$), in BEI by 27.0% ($p < 0.05$), and in event and ramp counts by 35.0% ($p < 0.05$) and by 29.0% ($p < 0.05$), respectively (Figure 1). Despite these improvements, all baroreflex activity parameters in Group A remained lower compared to the baseline values of Group C. On the contrary, Group B presented a small, but no significant reduction of the BRS and BEI values at the end of the study.

At the end of the study, exercise training resulted in a significant increase in VO₂peak by 22.4% ($p < 0.05$), ET by 23.8% ($p < 0.05$),

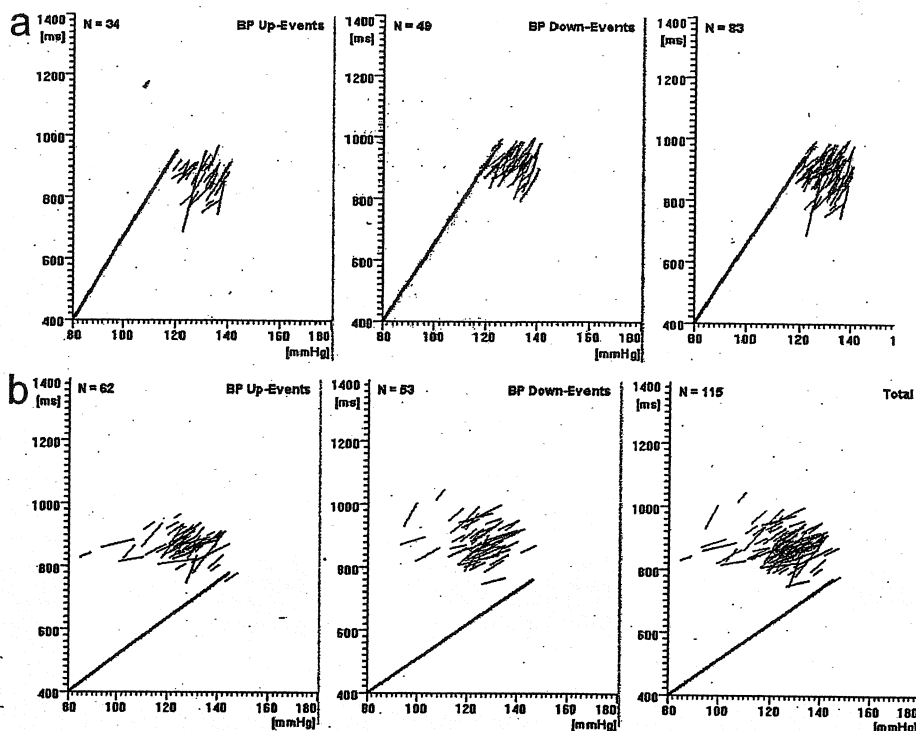


Figure 1. Baroreflex sensitivity course of a patient, during 30 min of the tilt test, pre (a) and post (b) training. Rising blood pressure and falling blood pressure events and mean slope of the baroreflex are shown.

METs by 22.4% ($p < 0.05$), VE_{peak} by 33.8% and HR_{peak} by 12.8% ($p < 0.05$) in Group A. On the contrary, the parameters of spiroergometric study in Group B were not significantly changed. At the end of the 7-month exercise program, the following correlations were observed in Group A:

- high correlations between event counts and ET ($r = 0.799$, $p < 0.01$), METs ($r = 0.736$, $p < 0.01$), VO_{2peak} ($r = 0.715$, $p < 0.01$) and VE_{peak} ($r = 0.699$, $p < 0.01$),
- weak correlations between BRS and METs ($r = 0.476$, $p < 0.05$), and VO_{2peak} ($r = 0.443$, $p < 0.05$), and
- weak correlations between BEI and METs ($r = 0.492$, $p < 0.05$) and VO_{2peak} ($r = 0.467$, $p < 0.05$).

Discussion

This study adds insights into the currently ongoing discussion dealing with BRS and exercise training programs in patients with chronic diseases. The main finding was the improvement of cardiac baroreflex activity with intradialytic training in HD patients. Ad-

ditionally, exercise during hemodialysis led to an improvement of patients' cardiorespiratory efficiency.

At baseline BRS and BEI were found to be reduced in our HD patients, compared to healthy individuals. This finding is clinically important since cardiac baroreceptors are known to have a crucial role in the short term control of blood pressure. Moreover, reduced BRS values were found to be associated with arrhythmias [Herzog 2003] and increased mortality in CKD patients [Herzog 2003, Johansson et al. 2007]. It is also interesting that our patients achieved a markedly low number of ramp and event counts during BRS measurements. Similar results in CKD patients were obtained by other investigators [Johansson et al. 2005, Pickering et al. 1972]. Additionally, Tomiyama et al. [1980] observed severely depressed BRS in predialysis patients compared with healthy individuals. In addition, the values of BRS found in our patients were similar to these of patients with other chronic diseases such as hypertension, diabetes mellitus, myocardial infarction, chronic obstructive pulmonary disease and coronary artery disease [Costes et al. 2004,

Iellamo et al. 2000, La Rovere et al. 1998, Loimala et al. 2003, Pikkujamsa et al. 1998].

There is a variety of mechanisms that could be responsible for the reduced BEI and BRS in HD patients, since the baroreflex activity is modulated both centrally and peripherally. Studies have shown that cardiac vagal motor neurons are affected by areas located in the prefrontal cortex and the insular cortex [Davos et al. 2002]. It was supported that ischemic changes affecting cortical areas which are critical for the cardiovascular control contribute to the reduced BEI in HD patients [Davos et al. 2002]. It was also demonstrated that in chronic HD patients, reduced BRS is associated with arterial stiffness due to vascular calcification [Chesterton et al. 2005]. Additionally, vascular hypertrophy resulting from hypertension seems to affect the BRS [Johansson et al. 2005]. The abnormality of baroreceptor response in HD patients could also be explained by the increased sympathetic activity and uremic neuropathy which are common in the majority of the patients [Johansson et al. 2005, Robinson and Carr 2002]. Nevertheless, no correlations were found between BRS and duration of dialysis or kidney disease in our patient groups. However, a significant result of our study was the strong correlation between aerobic capacity and BRS. This supports the supposition that cardiorespiratory insufficiency could contribute to the low values of baroreflex activity in HD patients.

At baseline, cardiorespiratory capacity indices of our HD patients were found to be markedly impaired compared with the healthy controls; as was also mentioned by other reports [Altintepe et al. 2006, Konstantinidou et al. 2002, Koudi et al. 1998, Sterky and Stegmayr 2005]. Exercise training during hemodialysis had beneficial effect on exercise tolerance. Our results indicate that a 7-month intradyalitic training program increases significantly both $\text{VO}_{2\text{peak}}$ and exercise capacity. It is already well-documented that the improvement of aerobic capacity after training is mainly caused by the central and peripheral adaptations to exercise [Koudi et al. 1998, Painter et al. 1986].

The main finding of this study is the improvement of cardiac baroreflex activity following exercise training in HD patients. However, the BRS values still remained lo-

wer compared to those of healthy sedentary controls. Similar improvements were found previously after participation in an exercise rehabilitation program in patients with coronary artery disease, COPD and diabetes Type 2, as well as in elderly subjects [Costes et al. 2004, Iellamo et al. 2000, Pichot et al. 2005]. Moreover, it was supported that nocturnal hemodialysis increases BRS via greater afferent baroreceptor responsiveness to pulsatile pressure [Chan et al. 2005].

However, there are studies that have demonstrated no effects of short-term exercise training on baroreflex sensitivity in healthy individuals [Bowman et al. 1997, Sheldahl et al. 1994]. Evidently long-term exercise training is needed to obtain a baroreflex gain in healthy people. Moreover, it is possible that individuals with poor initial exercise tolerance and depressed BRS have more chances to improve significantly the baroreflex activity with exercise [Pichot et al. 2005, Ueno and Moritani 2003].

It seems plausible that the mechanisms by which exercise training increases the BRS might include improvement of endothelial function, increase of endoneurial blood flow and enhancement in vessel wall properties [Wolf et al. 2000]. Besides, bradycardia, a direct consequence of exercise, is known to give rise to an improved baroreflex response [Chapleau et al. 1989]. Improved BRS by means of exercise may also be related to repeated activation of baroreflex. During physical exercise, rapid changes in systemic resistance and blood pressure activate the vascular wall [Loimala et al. 2003]. Thus, exercise training may also "train" the baroreflex to respond more effectively to blood pressure changes. This mechanism is confirmed by our results since the event and ramp counts were found to be significantly increased at the end of the study in the trained group. This means that after exercise training the baroreflex is more sensitive in arterial pressure changes.

Interestingly, significant relationships were observed between BRS, BEI, exercise time and METs following 7 months of training. Similar positive relationship between physical activity and BRS was noticed in healthy population after exercise training [Kardos et al. 2001]. Moreover, enhanced BRS, according to improvement of functional ca-

capacity, has been noticed in patients with diabetes Type 2, coronary artery disease, myocardial infarction and chronic obstructive pulmonary disease following exercise training [Costes et al. 2004, Iellamo et al. 2003, La Rovere et al. 1998, Loimaala et al. 2003]. Thereby, these findings support the hypothesis that individuals with better functional capacity have also greater baroreflex function.

In conclusion, exercise training during dialysis improved the baroreflex sensitivity in combination with the cardiorespiratory and work capacity benefits. However, additional studies need to examine if these gains decrease the cardiovascular morbidity and mortality in HD patients.

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